AMPLENESS IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites items from Divisors on Spaces, Topologies on Spaces, Morphisms of Spaces, Cohomology of Spaces and Morphisms, plus same-chapter descent lemmas (quasi-compact, separated, open immersion), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Base change of relatively ample modules. Stacks Project, Divisors on Spaces, Lemma spaces-divisors-lemma-ample-base-change. Proves (1) implies (2).
- Relative ampleness is \'etale local on the base. Stacks Project, Divisors on Spaces, Lemma spaces-divisors-lemma-relatively-ample-local. Lets the proof assume Y is a scheme.
- Refining fpqc coverings by schemes. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-refine-fpqc-schemes. Lets the proof assume each Y_i is a scheme.
- Properties of relatively ample morphisms. Stacks Project, Divisors on Spaces, Lemma spaces-divisors-lemma-relatively-ample-properties. Gives representability, quasi-compactness and separatedness of $X_i \to Y_i$, and the final ampleness characterization.
- Pushforward of quasi-coherent sheaves. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-pushforward. Shows $\mathcal{A}$ is a quasi-coherent graded $\mathcal{O}_Y$-algebra.
- Flat base change of pushforwards. Stacks Project, Cohomology of Spaces, Lemma spaces-cohomology-lemma-flat-base-change-cohomology. Shows the formation of $\mathcal{A}$ commutes with flat base change, so $\mathcal{A}_i = g_i^*\mathcal{A}$.
- Map into relative Proj from an invertible module. Stacks Project, Divisors on Spaces, Lemma spaces-divisors-lemma-invertible-map-into-relative-proj. Supplies the open subset $U(\psi)$ and its behavior under base change.
- Characterization of relatively ample invertible modules. Stacks Project, Morphisms, Lemma morphisms-lemma-characterize-relatively-ample. Supplies the stalk surjectivity criterion and the Proj open-immersion criterion.
- fpqc descent of open immersions. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-open-immersion. Descends the open immersion $r_{\mathcal{L}, \psi}$ from the Y_i.

Source reference. Original reference 0D33: sources/stacks-chow/tags.txt, source lines 16403–16403. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0D36: sources/stacks-chow/tags.txt, source lines 16406–16406. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0419: sources/stacks-chow/tags.txt, source lines 5284–5284. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D34: sources/stacks-chow/tags.txt, source lines 16404–16404. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041L: sources/stacks-chow/tags.txt, source lines 5296–5296. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0421: sources/stacks-chow/tags.txt, source lines 5312–5312. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03M9: sources/stacks-chow/tags.txt, source lines 4741–4741. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 073K: sources/stacks-chow/tags.txt, source lines 9064–9064. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01VJ: sources/stacks-chow/tags.txt, source lines 2448–2448. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D2Z: sources/stacks-chow/tags.txt, source lines 16399–16399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041X: sources/stacks-chow/tags.txt, source lines 5308–5308. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $\{g_i : Y_i \to Y\}_{i \in I}$ be an fpqc covering. Let $f_i : X_i \to Y_i$ be the base change of f and let $\mathcal{L}_i$ be the pullback of $\mathcal{L}$ to X_i. The following are equivalent*

(1) *$\mathcal{L}$ is ample on X/Y, and*
(2) *$\mathcal{L}_i$ is ample on X_i/Y_i for every $i \in I$.*

Proof. The implication (1) $\Rightarrow$ (2) follows from Divisors on Spaces, Lemma 0D33. Assume (2). To check $\mathcal{L}$ is ample on X/Y we may work étale locally on Y, see Divisors on Spaces, Lemma 0D36. Thus we may assume that Y is a scheme and then we may in turn assume each Y_i is a scheme too, see Topologies on Spaces, Lemma 0419. In other words, we may assume that $\{Y_i \to Y\}$ is an fpqc covering of schemes.

By Divisors on Spaces, Lemma 0D34 we see that $X_i \to Y_i$ is representable (i.e., X_i is a scheme), quasi-compact, and separated. Hence f is quasi-compact and separated by Lemmas 041L and 0421. This means that $\mathcal{A} = \bigoplus_{d \geq 0} f_* \mathcal{L}^{\otimes d}$ is a quasi-coherent graded $\mathcal{O}_Y$-algebra (Morphisms of Spaces, Lemma 03M9). Moreover, the formation of $\mathcal{A}$ commutes with flat base change by Cohomology of Spaces, Lemma 073K. In

particular, if we set $\mathcal{A}_i = \bigoplus_{d\geq 0} f_{i,*}\mathcal{L}_i^{\otimes d}$ then we have $\mathcal{A}_i = g_i^*\mathcal{A}$. It follows that the natural maps $\psi_d : f^*\mathcal{A}_d \to \mathcal{L}^{\otimes d}$ of $\mathcal{O}_X$ pullback to give the natural maps $\psi_{i,d} : f_i^*(\mathcal{A}_i)_d \to \mathcal{L}_i^{\otimes d}$ of $\mathcal{O}_{X_i}$-modules. Since $\mathcal{L}_i$ is ample on X_i/Y_i we see that for any point $x_i \in X_i$, there exists a $d \geq 1$ such that $f_i^*(\mathcal{A}_i)_d \to \mathcal{L}_i^{\otimes d}$ is surjective on stalks at x_i. This follows either directly from the definition of a relatively ample module or from Morphisms, Lemma 01VJ. If $x \in |X|$, then we can choose an i and an $x_i \in X_i$ mapping to x. Since $\mathcal{O}_{X,\overline{x}} \to \mathcal{O}_{X_i,\overline{x}_i}$ is flat hence faithfully flat, we conclude that for every $x \in |X|$ there exists a $d \geq 1$ such that $f^*\mathcal{A}_d \to \mathcal{L}^{\otimes d}$ is surjective on stalks at x. This implies that the open subset $U(\psi) \subset X$ of Divisors on Spaces, Lemma 0D2Z corresponding to the map $\psi : f^*\mathcal{A} \to \bigoplus_{d\geq 0} \mathcal{L}^{\otimes d}$ of graded $\mathcal{O}_X$-algebras is equal to X. Consider the corresponding morphism

$$r_{\mathcal{L},\psi} : X \longrightarrow \underline{\mathrm{Proj}}_Y(\mathcal{A})$$

It is clear from the above that the base change of $r_{\mathcal{L},\psi}$ to Y_i is the morphism $r_{\mathcal{L}_i,\psi_i}$ which is an open immersion by Morphisms, Lemma 01VJ. Hence $r_{\mathcal{L},\psi}$ is an open immersion by Lemma 041X. Hence X is a scheme and we conclude $\mathcal{L}$ is ample on X/Y by Morphisms, Lemma 01VJ. □